React Query Options Explained

One-line definitions with simple explanations for the most common `useQuery` and `QueryClient` options.

Caching and freshness

`staleTime: 1000 * 60 * 5`

- **Definition:** How long fetched data counts as "fresh" (no refetch needed).
- **Simple:** Think of milk with a 5-minute freshness stamp. For 5 minutes React Query trusts the cache and makes no new request. After that the data is "stale" and will refetch on the next trigger. The default is `0`, so data is stale immediately.

`gcTime: 1000 * 60 * 10` (formerly `cacheTime`)

- **Definition:** How long **unused** cached data stays in memory before it is deleted.
- **Simple:** The recycling bin timer. When no component is using a query, its data stays for 10 minutes. If the user comes back within that time, it shows instantly. After that it is deleted. The default is 5 minutes.

> `staleTime` is about when to refetch. `gcTime` is about when to delete from memory.

Retry

`retry: 3`

- **Definition:** How many times to retry a failed request before showing an error.
- **Simple:** If the call fails, try 3 more times before giving up. The default is 3. Use `false` for no retries.

`retryDelay: (attemptIndex) => Math.min(1000 * 2 ** attemptIndex, 30000)`

- **Definition:** How long to wait between retries.
- **Simple:** Wait longer after each failure (exponential backoff): 1s, 2s, 4s, 8s, and so on, but never more than 30 seconds (`30000`).

Automatic refetch triggers

`refetchOnWindowFocus: false`

- **Definition:** Whether to refetch stale data when the user returns to the browser tab.
- **Simple:** The default is `true`, so switching tabs and coming back refreshes data. `false` turns that off.

`refetchOnMount: true`

- **Definition:** Whether to refetch when a component using the query mounts.
- **Simple:** `true` refetches only if the data is stale. `'always'` refetches every time. `false` never refetches on mount.

`refetchOnReconnect: 'always'`

- **Definition:** Whether to refetch when the internet connection comes back.
- **Simple:** `'always'` refetches after reconnecting even if the data is fresh. `true` refetches only if stale. `false` never does.

Conditional fetching and polling

`enabled: !!userId`

- **Definition:** Turns the query on or off.
- **Simple:** `!!userId` turns the value into true or false. If `userId` is missing, the query waits. When `userId` appears, it runs. This is also called a dependent query.

`refetchInterval: false` (or milliseconds)

- **Definition:** Refetches automatically on a timer (polling).
- **Simple:** `5000` refetches every 5 seconds, which is useful for live data like a dashboard. `false` means no polling.

`refetchIntervalInBackground: false`

- **Definition:** Whether polling continues when the browser tab is not in focus.
- **Simple:** `false` pauses polling in a hidden tab to save requests. `true` keeps polling anyway.

Data shaping and placeholders

`select: (data) => data.user`

- **Definition:** Transforms or picks part of the data before the component gets it.
- **Simple:** The API returns a big object, but you only want `user`. The cache keeps the full response, and the component receives only `data.user`. It is also useful for filtering or sorting.

`placeholderData: { name: 'Loading...' }`

- **Definition:** Temporary fake data shown while the real data loads.
- **Simple:** A mannequin in the shop window until the real item arrives. It is **not saved to the cache**, and the query is still in `pending` state, so a real fetch still happens.

`placeholderData: keepPreviousData`

- **Definition:** A ready-made helper that keeps showing the previous query's data while the new key loads.
- **Simple:** Used in pagination. When you click Next, page 1 stays visible until page 2 arrives, so the list doesn't flicker or go blank. This is the same option as the one above, just with a different value.

`initialData: undefined`

- **Definition:** Pre-fills the cache with real data before any fetch.
- **Simple:** You already have the data, for example from a list, so you put it straight into the cache. **It is saved to the cache** and treated as real data. With `staleTime` above 0, React Query may skip the fetch entirely. `undefined` means nothing is pre-filled.

`placeholderData` vs `initialData`

	`placeholderData`	`initialData`
Saved in cache	No	Yes
Treated as real data	No	Yes
Query status while showing it	`pending` (still loading)	`success`
Real fetch still happens	Yes	Depends on `staleTime`

Quick summary table

Option	Remember it as
`staleTime`	How long data stays fresh
`gcTime`	How long unused data stays in memory
`retry`	Number of retry attempts
`retryDelay`	Wait time between retries
`refetchOnWindowFocus`	Refetch when the tab is focused
`refetchOnMount`	Refetch when the component mounts
`refetchOnReconnect`	Refetch when the internet returns
`enabled`	Switch the query on or off
`refetchInterval`	Polling timer
`refetchIntervalInBackground`	Poll even in a hidden tab
`select`	Pick or transform the data
`placeholderData`	Temporary data, not cached
`initialData`	Real starting data, cached